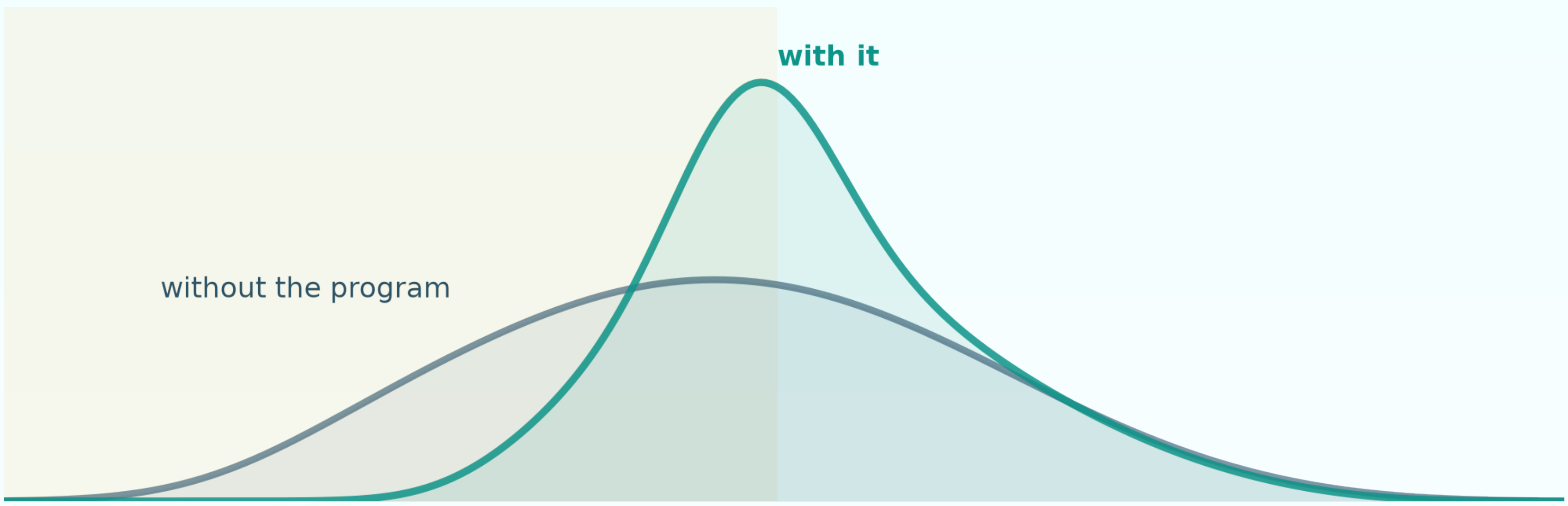
diff - diff

The effect your DiD

can't see.



Real programs move parts of the distribution.

The mean averages them away.

ChangesInChanges: distributional DiD. Now in diff-diff.

Athey & Imbens (2006), Econometrica 74(2), 431-497.

THE PROBLEM

A real effect,

averaged into nothing.

A loyalty program lifts low spenders by up to 65% and top spenders not at all. (Simulated customer-spend 2x2, so the truth is known by construction.) Run the standard playbook:

$0.22

mean DiD, in dollars

p = 0.90   CI [-$3.18, $3.62]

verdict: kill the program

$3.01

the TRUE mean effect

known by construction

The effect lives where

the mean isn't looking.

And it gets worse.

THE TWIST

Same data. One change:

log(spend).

+14.0%

p = 0.001

Your scale choice just became

your conclusion.

And neither verdict is right (truth: $3.01). The simulated market trend is a power-law transform of spend - additive on neither scale across groups with different customer mixes.

Choosing a scale IS choosing the assumption.

THE IDEA

Athey & Imbens (2006):

compare ranks, not scales.

"

The assumptions of the proposed model are invariant to the scaling of the outcome.

- Athey & Imbens (2006), Econometrica 74(2), p. 431

Their Changes-in-Changes model swaps parallel trends in means for a rank story: a treated customer's counterfactual is the outcome move of a control customer at the same rank. No scale enters the assumptions - and none enters the answer.

The receipt (median counterfactual quantile):

$20.3166	**$20.3166**
fit in dollars	exp(fit in logs)

Identical across all 19 quantiles - max relative difference 0.00e+00 on this run (unconditional fits; means and ATTs stay scale-specific for every estimator).

THE MATH

One composition builds the whole counterfactual.

$$F_{11}^{N}(y) \;=\; F_{10}\left(F_{00}^{-1}\left(F_{01}(y)\right)\right)$$

3. how many treated-pre customers sit below it

2. the same rank's outcome in control-pre

1. the rank of y among control-post outcomes

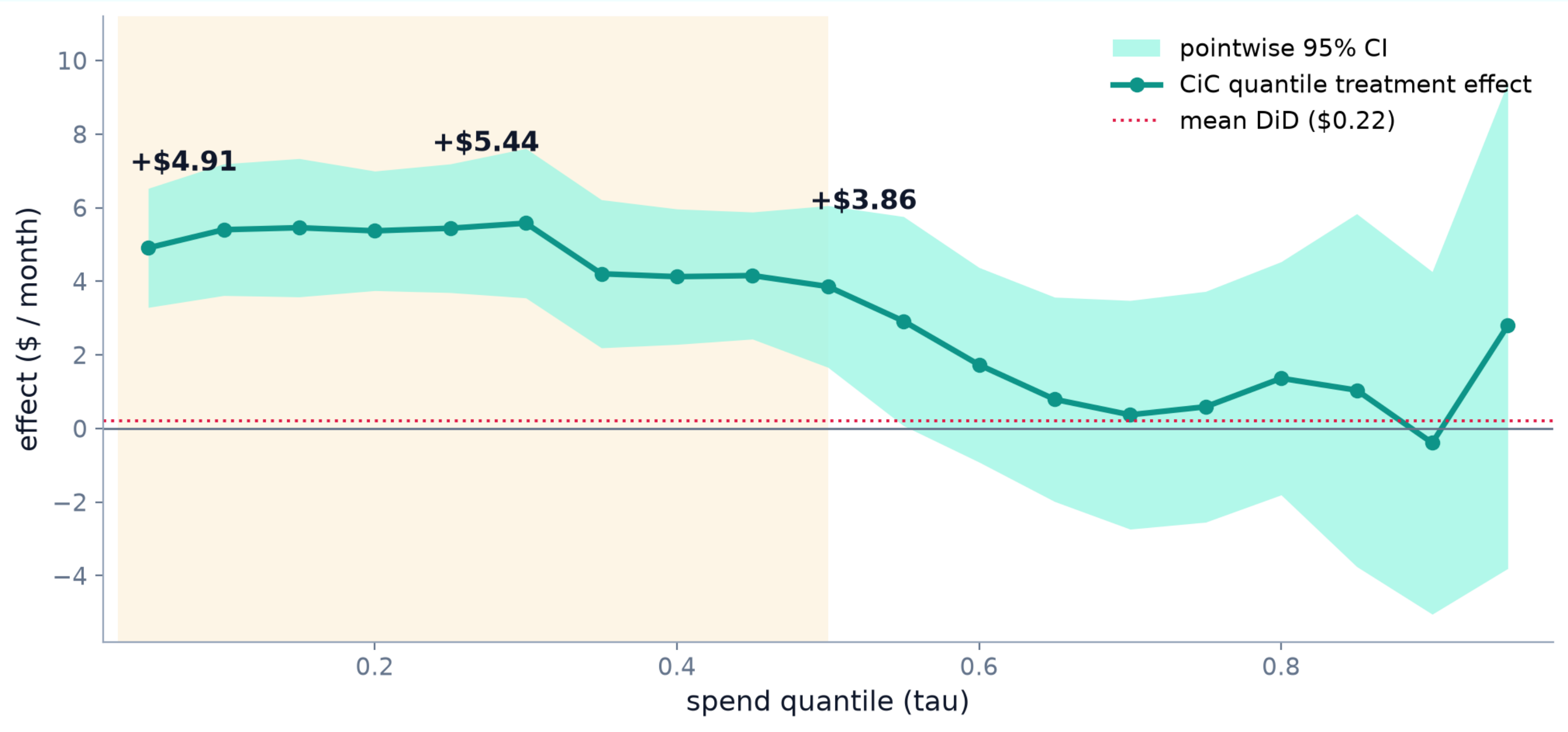
$$\mathrm{QTE}(\tau) \;=\; F_{11}^{-1}(\tau) \;-\; F_{11}^{N,\,-1}(\tau)$$

Read inside-out: rank a treated outcome among control-post, walk that rank back to control-pre, and count the treated-pre customers below it. The result is the treated group's entire no-program outcome distribution (Theorem 3.1) - quantile treatment effects at every tau, and the ATT as its mean. Nonparametrically identified; panel or repeated cross-sections.

THE PAYOFF

Same program.

Now visible.

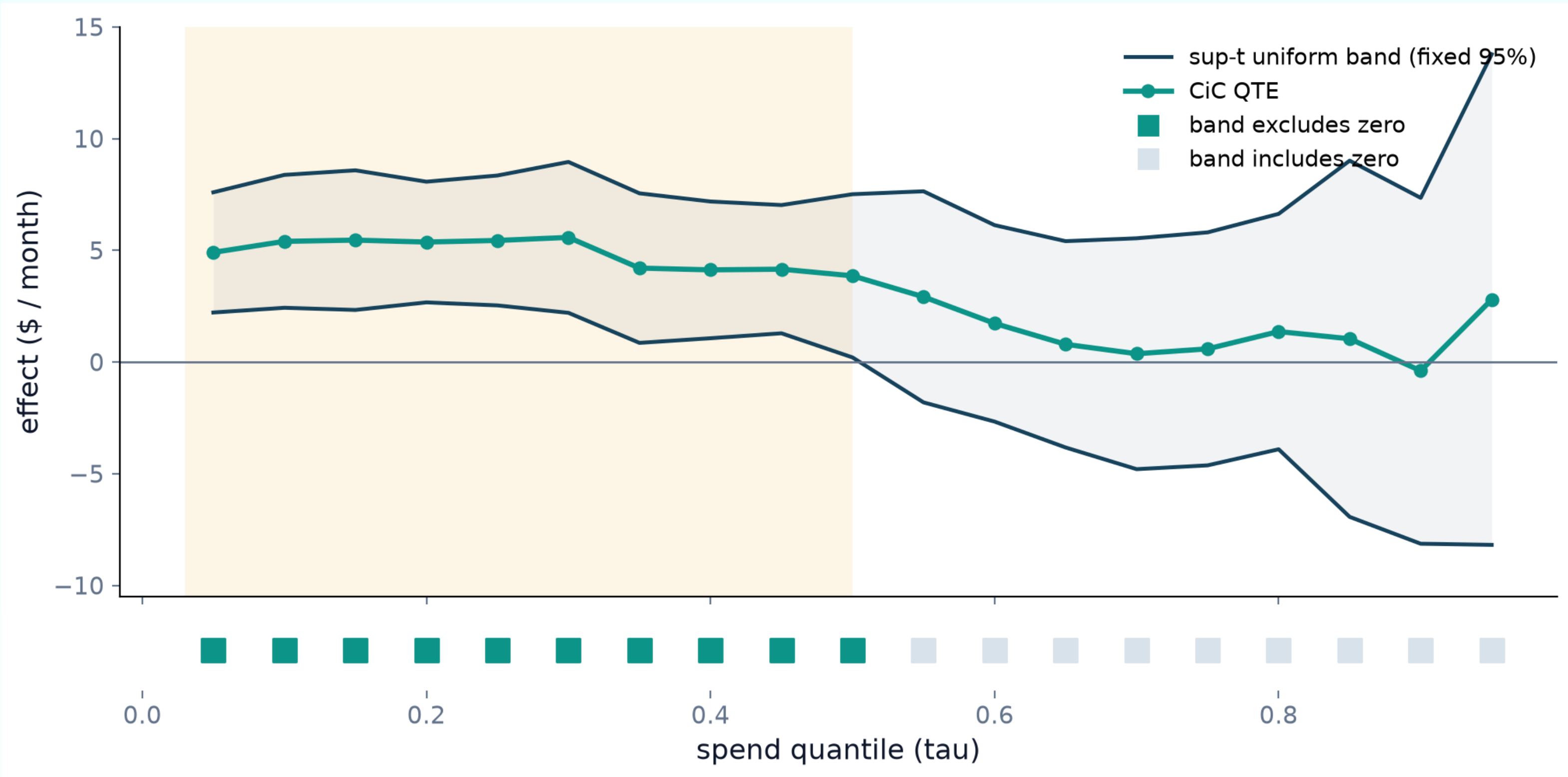


Gains of about +$5/month across the bottom half of the spend distribution, fading to nothing distinguishable from zero above the 70th percentile. And the ATT - $3.05 (p = 0.004) - lands on the $3.01 truth the mean missed.

ONE CLAIM

"The bottom half moved."

One test, nineteen quantiles.



Nineteen pointwise intervals over-reject when read together. Sup-t uniform bands (fixed 95%, a qte-parity convention) exclude zero for every quantile from 0.05 through 0.50 - and none above. The pointwise blip at tau = 0.55 (p = 0.044) evaporates under the bands: exactly the trap they exist to prevent. Above the median the bands are silent, not exonerating.

THE CODE

Three lines to the whole distribution.

```
from diff_diff import ChangesInChanges

results = ChangesInChanges(n_bootstrap=999, seed=27).fit(
    df, outcome='spend', treatment='treated', time='post',
)

results.att                 # $3.05 (p = 0.004)
results.quantile_effects    # QTE at 19 quantiles
results.uniform_bands()     # sup-t joint bands

# customer mix differs?  fit(..., covariates=['tenure'])
# same units both periods?  ChangesInChanges(panel=True)
```

Same sklearn-style API as every diff-diff estimator.

COMPOSITION

When the mix differs, condition.

If the treated group's mix differs on something that also drives the trend, the unconditional comparison is confounded. Here: treated customers skew longer-tenure, tenure drives engagement growth, and the true effect is 6.0 points - for everyone.

9.38

unconditional CiC

CI [8.32, 10.45]

excludes the truth:

confidently wrong

6.53

covariates=['tenure']

CI [5.35, 7.70]

covers the truth (6.0)

The conditional fit compares within covariate values: per-cell linear quantile regressions on a 99-tau grid, following R qte's xformla conventions verbatim - with a conditional support diagnostic standing guard.

PRODUCTION-READY

Built for real pipelines.

Panel + repeated cross-sections

panel=True enforces balanced units + unit-block bootstrap

Bootstrap + sup-t uniform bands

seeded, reproducible; fixed-95% bands (qte parity)

Covariates

quantile-regression conditioning via covariates=[...]

Loud guardrails

support, interior range + ties warnings; NaN, never a guess

practitioner_next_steps()

distributional next-step guidance, built in

sklearn-style API

fit() / summary() / to_dataframe(), like every estimator

Validated against R qte 1.3.1

Unconditional point estimates match the qte golden files to 1e-10; the type-1 quantile arithmetic is pinned bit-exactly (atol=0). Covariate branch: conventions exact given R inputs; the LP solver is proven optimal at every tau.

diff - diff

Ask your 2x2 who moved -

not just how much on average.

```
pip install diff-diff
```

Want the full walkthrough? Tutorial 27 reproduces this example end to end:

diff-diff.readthedocs.io/en/latest/tutorials/27_cic_distributional_effects.html

github.com/igerber/diff-diff

ChangesInChanges - Athey & Imbens (2006). Now in diff-diff.